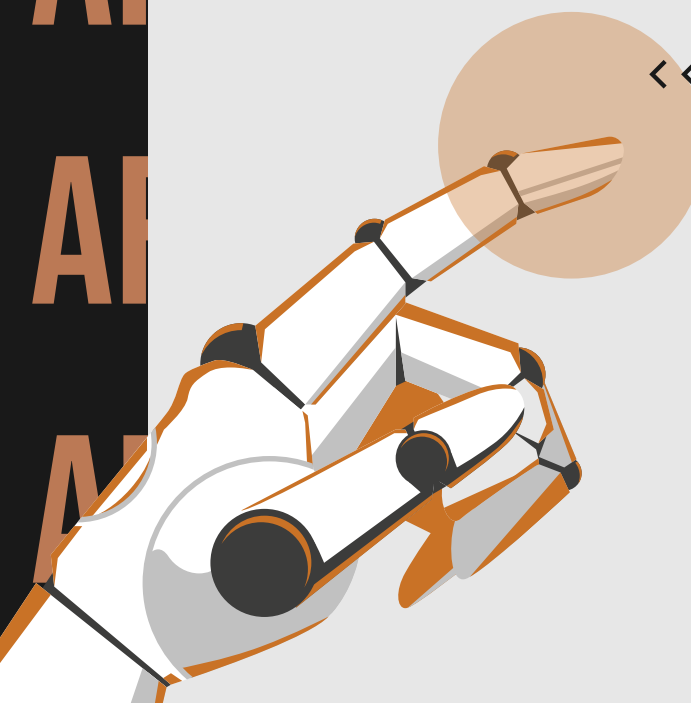
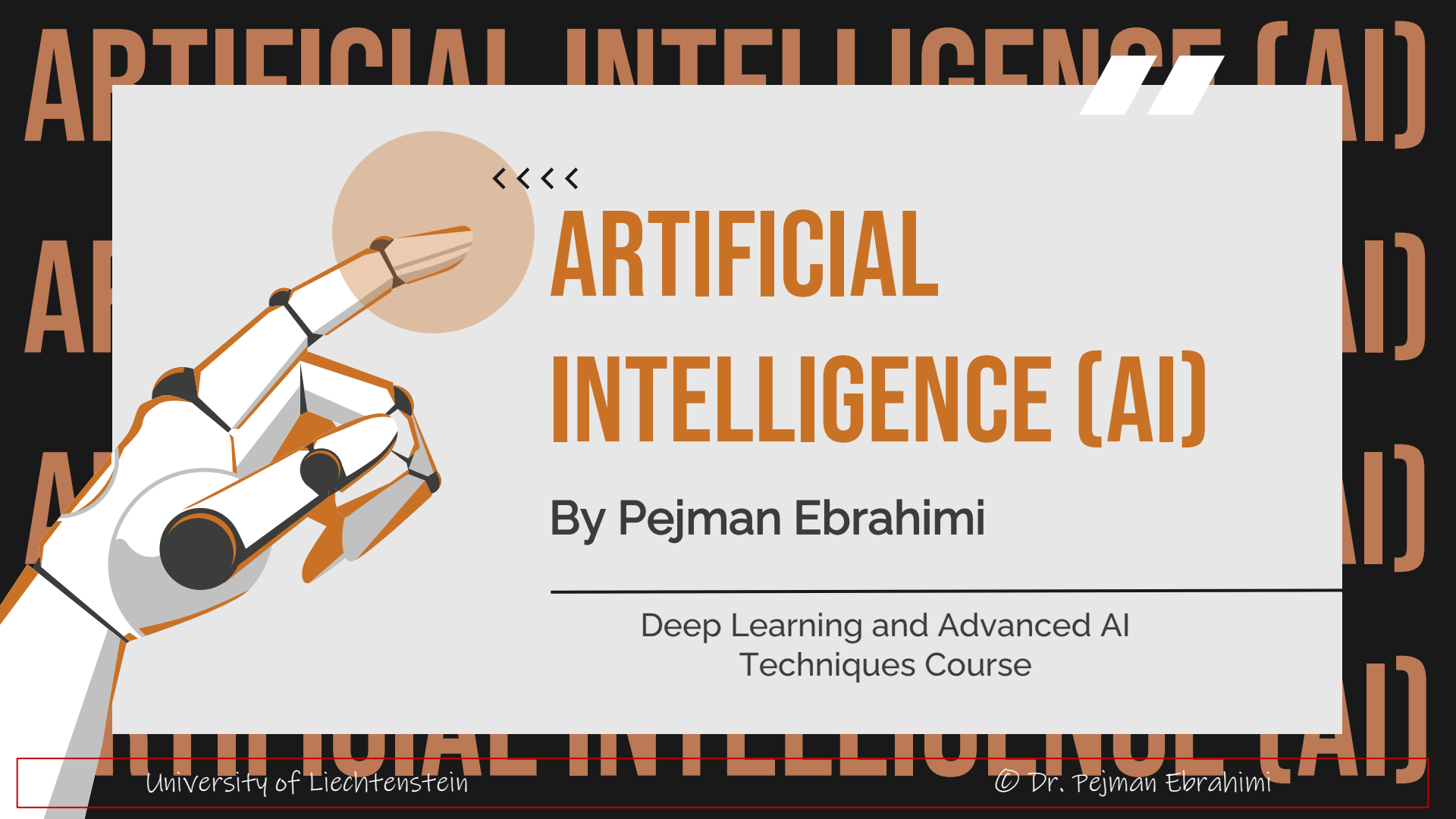




ARTIFICIAL INTELLIGENCE (AI)

By Pejman Ebrahimi

Deep Learning and Advanced AI
Techniques Course

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01.

RAG (RETRIEVAL-AUGMENTED GENERATION)



WHAT IS RAG?

❑ Definition:

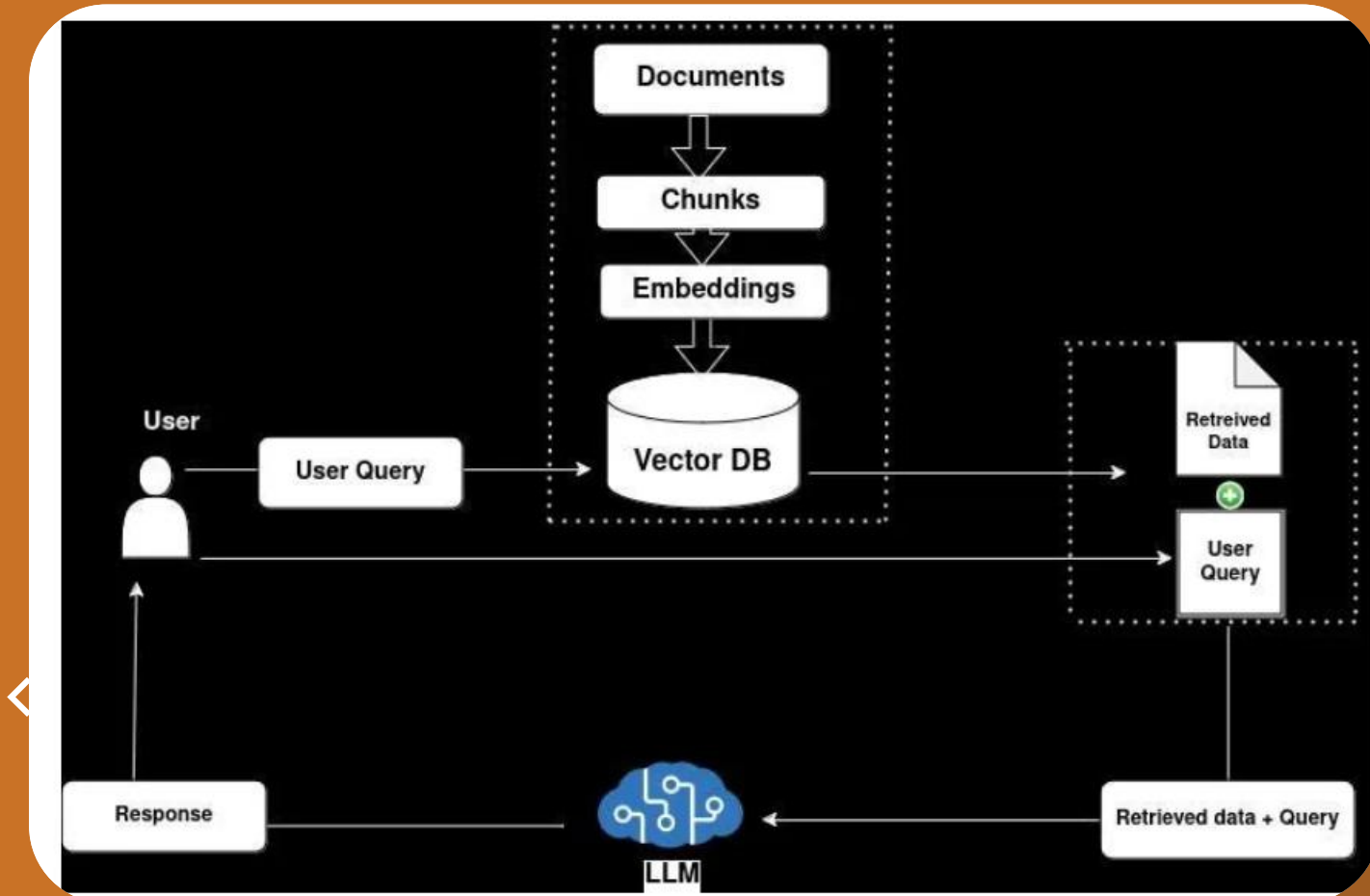
- RAG is an AI framework that combines a **retriever** (searches for relevant information) and a **generator** (creates responses) to answer questions more accurately and with up-to-date or domain-specific knowledge.

❑ Key Components:

- Retriever: Fetches documents or snippets from a knowledge base.
- Generator (LLMs models): Uses the retrieved content to generate an informed response.

❑ Why Use RAG?

- Improves factual accuracy Reduces hallucination Enables dynamic updates to knowledge without retraining the model.



HOW RAG WORKS?

- ❑ **Input:** User asks a question
- ❑ **Retrieval:** System searches a document store for relevant texts
- ❑ **Augmentation:** Retrieved texts are passed along with the question to the generator
- ❑ **Generation:** The model synthesizes a response using both the query and retrieved data

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ARTIFICIAL

INTELLIGENCE

(AI)

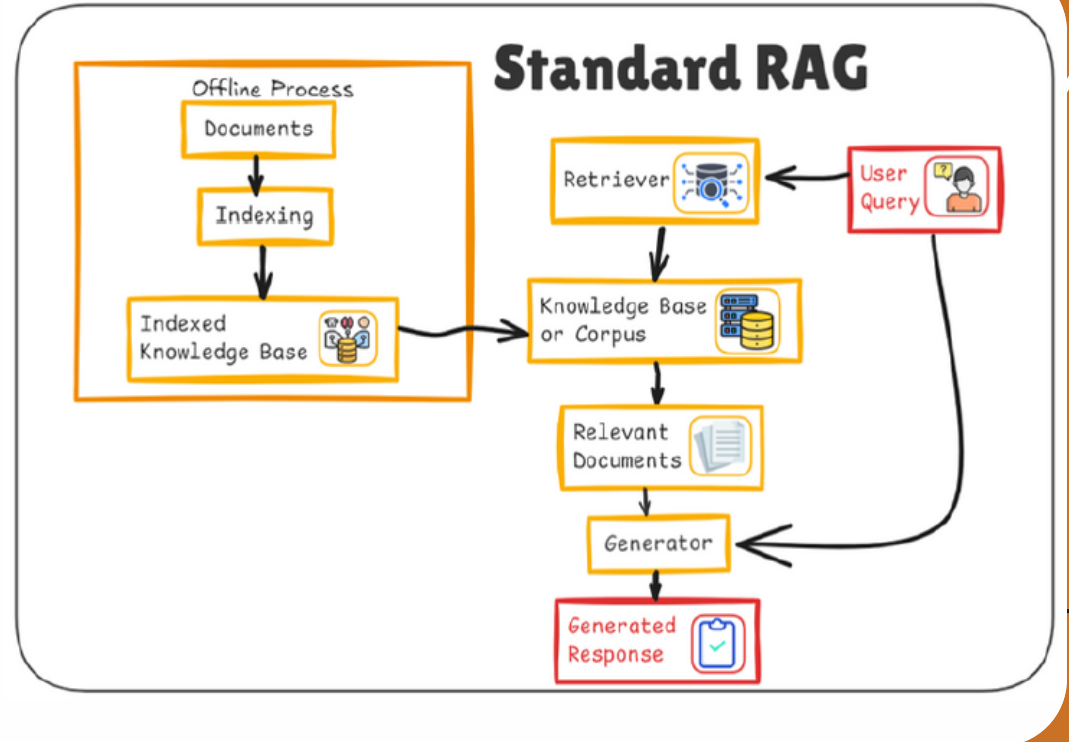
TYPES OF RAG



(AI)

Standard RAG

- Combines **retrieval** with **large language models** for accurate, context-aware responses.
- Breaks **documents into chunks** for efficient information retrieval.
- Aims for **1-2 second response times** for real-time use.
- **Enhances answer quality** by leveraging external data sources.



Corrective RAG

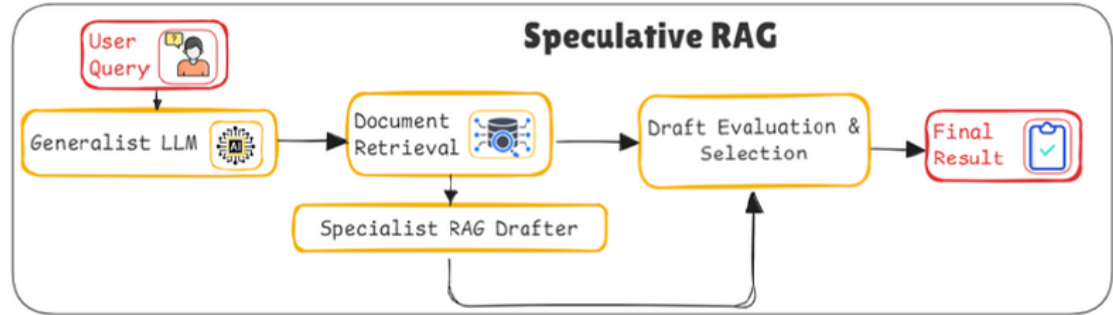


Corrective RAG

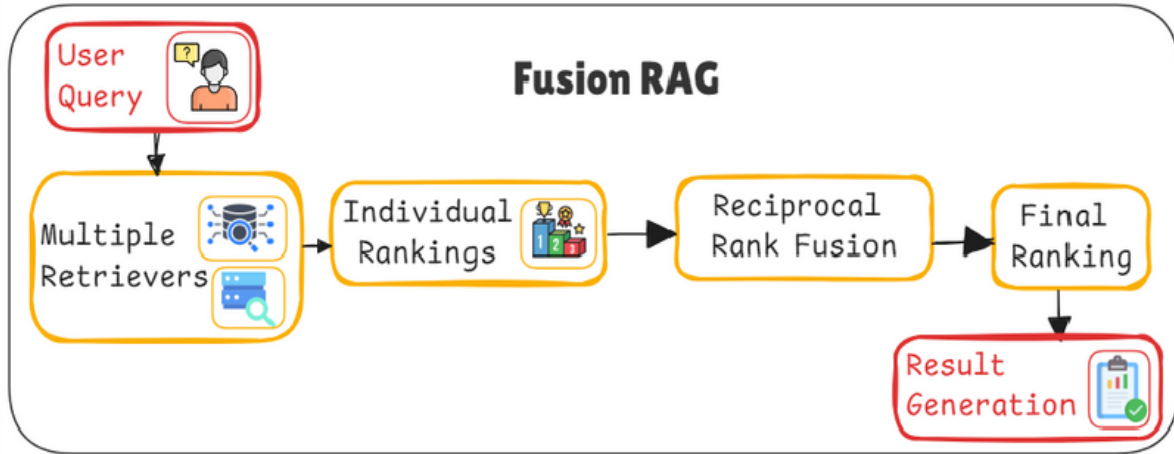
- Focuses on **identifying** and **fixing errors** in generated responses.
- Uses multiple passes to **improve outputs** based on feedback.
- Aims for **higher precision** and **user satisfaction** compared to standard RAG.
- Leverages user feedback to **enhance the correction** process.

Speculative RAG

- Uses a small **specialist model** for drafting and a larger **generalist model** for verification, ensuring efficiency and accuracy.
- **Parallel Drafting:** Speeds up responses by generating multiple drafts simultaneously.
- **Superior Accuracy:** Outperforms standard RAG systems.
- **Efficient Processing:** Offloads complex tasks to specialized models, reducing computational load.



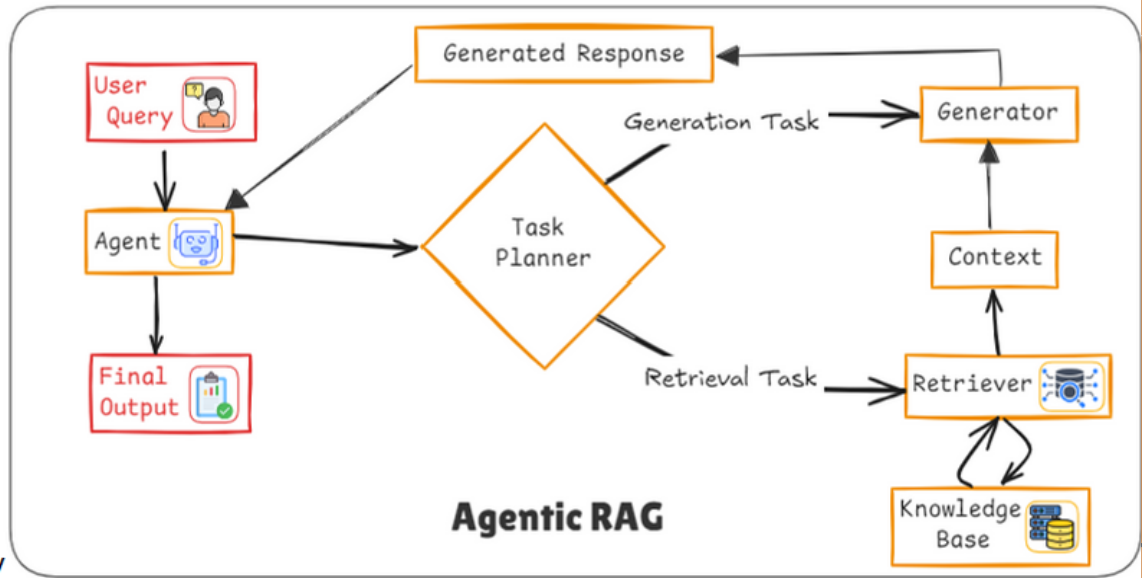
Fusion RAG



- Integrates **multiple** retrieval methods and data sources for enhanced response quality.
- Provides **comprehensive answers** by leveraging diverse data inputs.
- **Increases** system **resilience** by reducing dependence on a single source.
- Adapts retrieval **strategies dynamically** based on query context.

Agentic RAG

- Uses **adaptive agents** for real-time strategy adjustments in information retrieval.
- Accurately **interprets user intent** for relevant, trustworthy responses.
- **Modular design** enables easy integration of new data sources and features.
- Enhances **parallel processing** and **performance** on complex tasks by running agents concurrently.

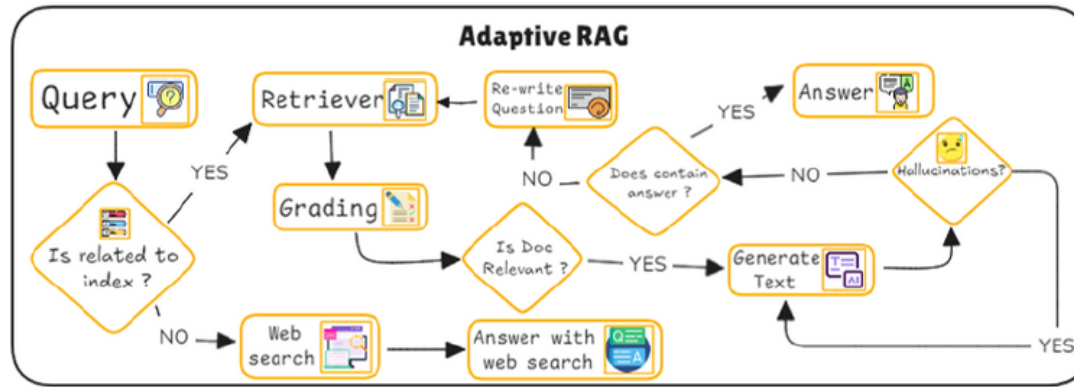


Self RAG



- Uses the model's own outputs as retrieval candidates for **better contextual relevance**.
- Refines responses iteratively, improving **consistency** and **coherence**.
- Grounds responses in prior outputs for **increased accuracy**.
- **Adapts retrieval** strategies based on the conversation's evolving context.

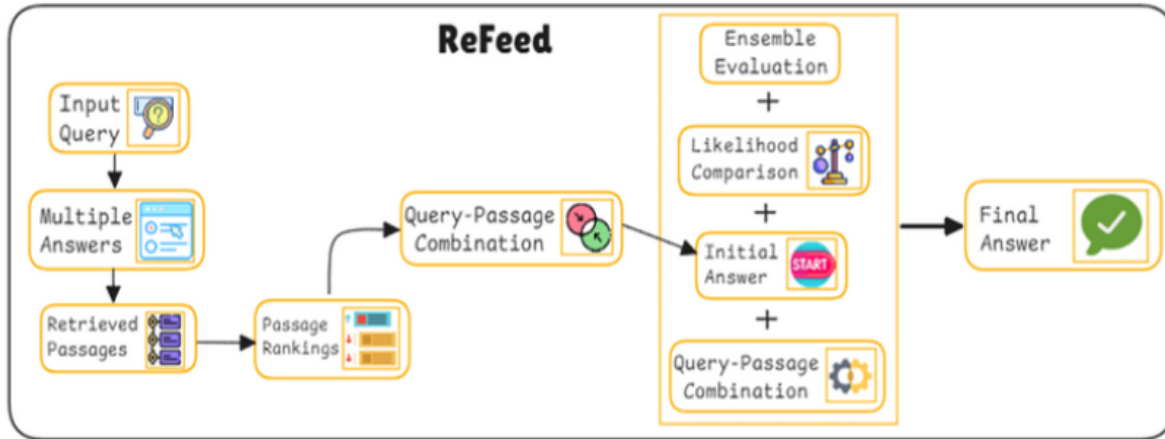
Adaptive RAG



- It **dynamically** decides when to retrieve external knowledge, balancing internal and external knowledge.
- It uses **confidence scores** from the language model's internal states to assess retrieval necessity.
- An honesty probe helps the model **avoid hallucinations** by aligning its output with its actual knowledge.
- It **reduces unnecessary retrievals**, improving both efficiency and response accuracy.

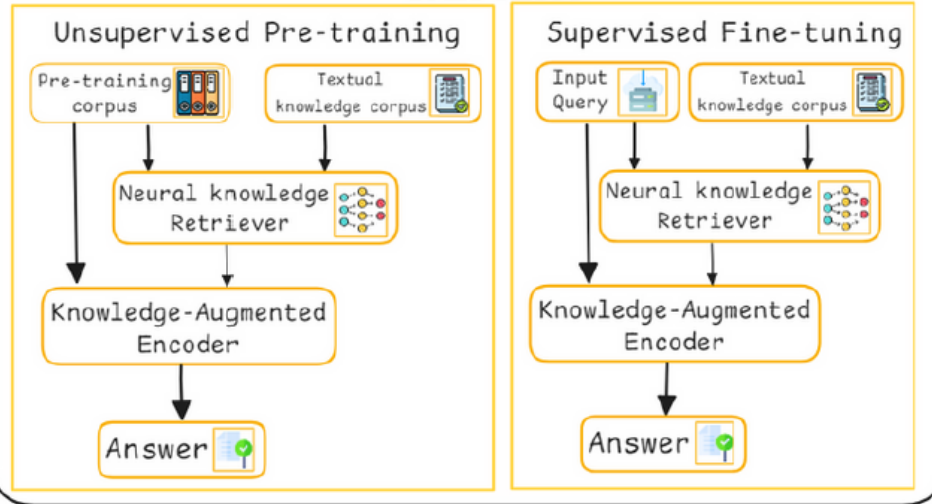
- REFEED refines model outputs using **retrieval feedback without fine-tuning**.
- Initial answers are improved by retrieving relevant documents and adjusting the response based on the new information.
- Generates **multiple answers** to improve retrieval accuracy.
- Combines pre- and post-retrieval outputs using a **ranking system to enhance answer reliability**.

REFEED Retrieval Feedback



REALM

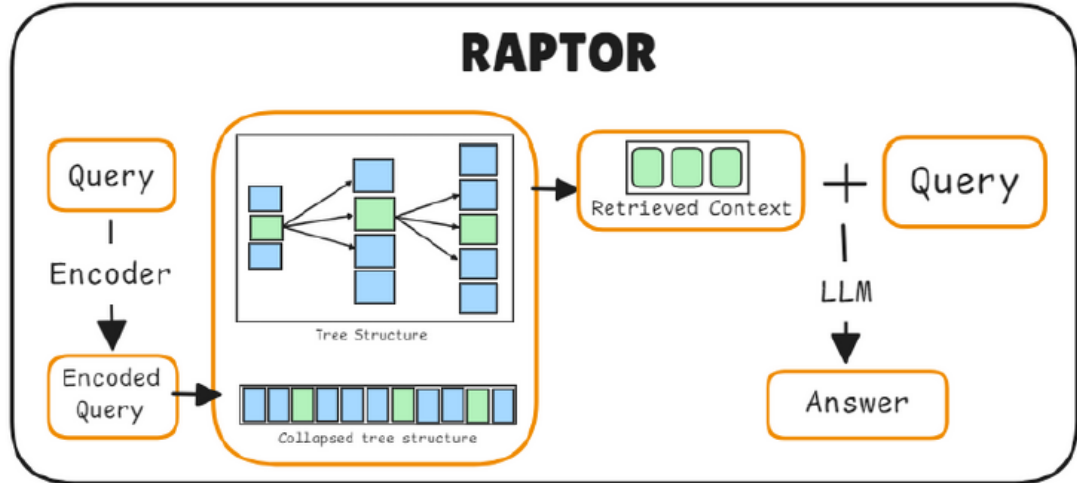
REALM



- REALM retrieves relevant documents from large corpora like Wikipedia to **enhance model predictions**.
- The retriever is trained with masked language modeling, optimizing retrieval to **improve prediction accuracy**.
- It uses **Maximum Inner Product Search** to efficiently find relevant documents from millions of candidates during training.
- REALM outperforms previous models in **Open-domain Question Answering** by integrating external knowledge.

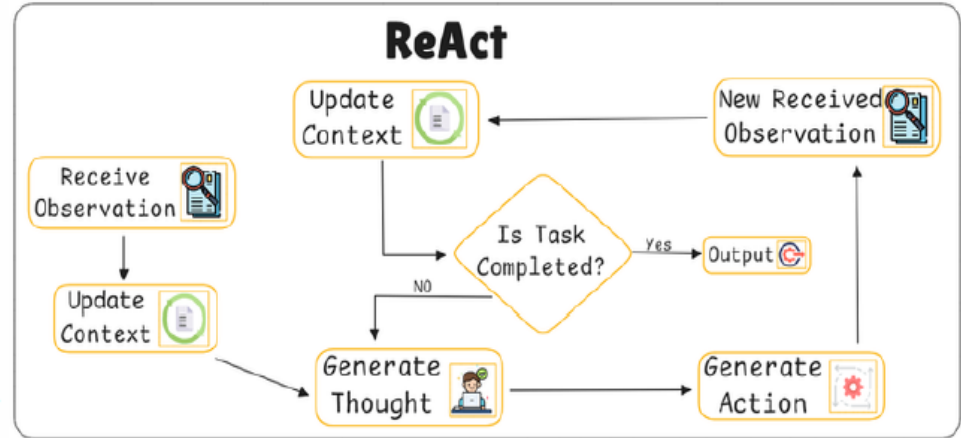
RAPTOR- Tree-Organized Retrieval

- RAPTOR builds a **hierarchical tree** by **clustering** and **summarizing text recursively**.
- It enables retrieval at **different abstraction levels**, combining **broad themes** with specific details.
- RAPTOR **outperforms traditional methods** in complex question-answering tasks.
- Offers tree traversal and collapsed tree methods for **efficient information retrieval**.



REACT

- The ReAct technique combines **reasoning and action**, allowing models to interact with their environment.
- It maintains situational **awareness by updating context** with past actions and thoughts.
- The model generates **task-aligned thoughts** to guide logical decision-making.
- **Real-time feedback** refines understanding, reducing errors and enhancing transparency and reliability.

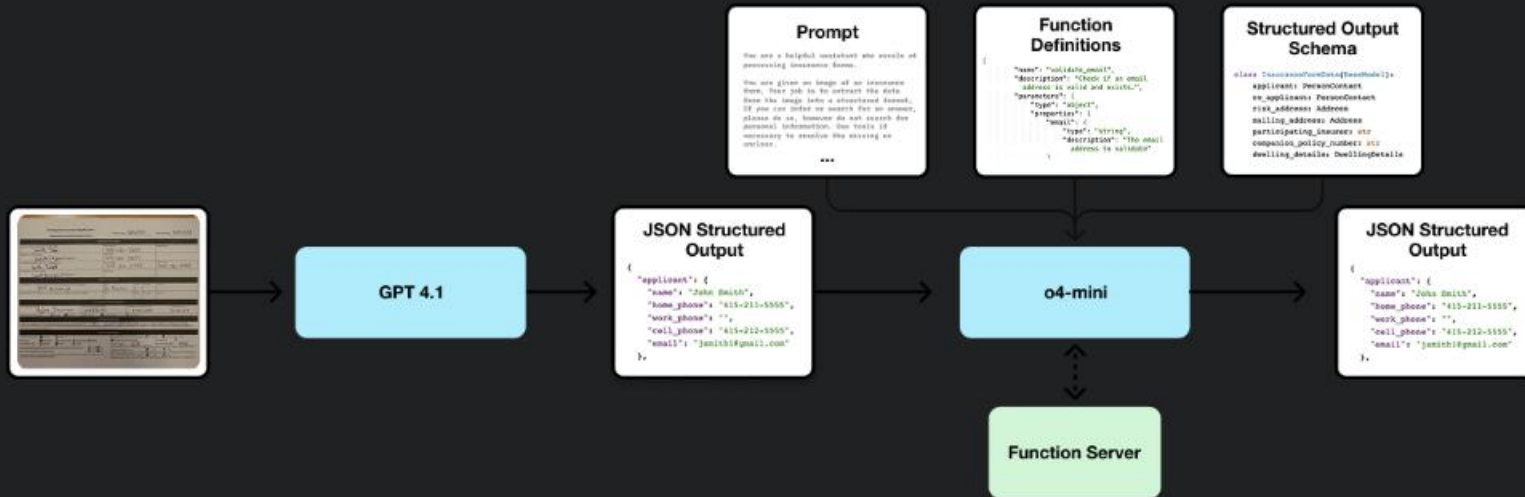


SOMETHING FROM FUTURE!

OPENAI'S INDEX-FREE LONG RAG



- That uses GPT-4.1 for retrieval without the need for dedicated index





02.

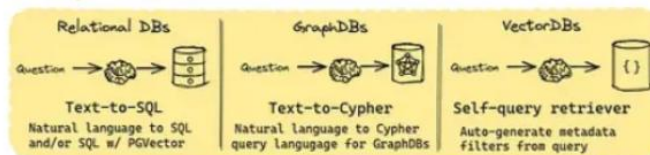
KAG (KNOWLEDGE-AUGMENTED GENERATION)



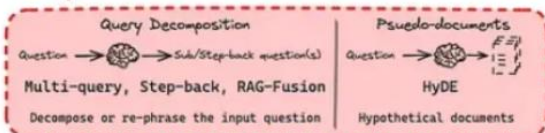
WHAT IS KAG?

- ❑ **KAG** or Knowledge-Augmented Generation presents a hybrid approach that enhances the generative capabilities of language models by directly incorporating structured knowledge graphs or external knowledge bases into the model architecture.
- ❑ Unlike RAG, which retrieves unstructured data, KAG focuses on the integration of structured knowledge to improve the quality of generation. It is based on the OpenSPG engine and addresses the limitations of traditional question-and-answer systems.

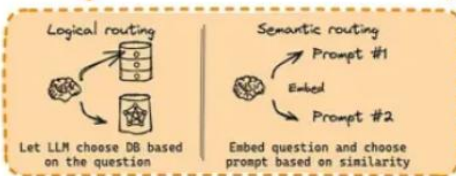
Query Construction



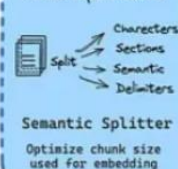
Query Translation



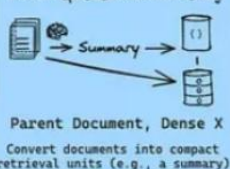
Routing



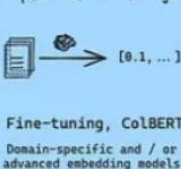
Chunk Optimization



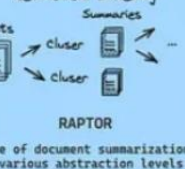
Multi-representation indexing



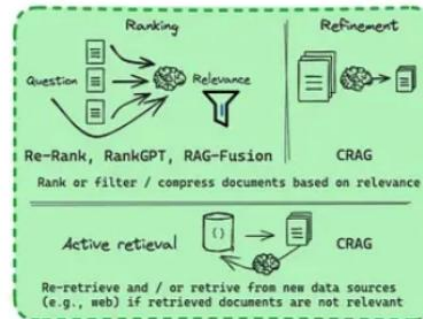
Specialized Embeddings



Heirarchical Indexing Summaries



Retrieval



Indexing

Generation



image by Lance Martin(Langchain)

HOW KAG WORKS?

- **What it is:** Similar to RAG but broader. KAG means any method where **external knowledge** (not just documents) is added to help generate better responses.
- **Types of knowledge:** Structured data (like knowledge graphs), ontologies, databases—not just text.
- **Why it's used:** To make responses more **factually correct, structured, or aligned with domain-specific knowledge**.
- **Example:** A medical chatbot using a **medical knowledge base** to give trustworthy health advice.

• Example: A medical chatbot using a medical knowledge base to give trustworthy health advice.

RAG & KAG

Feature	RAG	KAG
Data Source	Mostly unstructured text/docs	Structured and unstructured knowledge
Focus	Retrieving and generating	Injecting knowledge in various ways
Common Use Case	Chatbots, Q&A systems	Domain-specific assistants (e.g., legal, medical)
Knowledge Format	Text-based (documents, articles)	Graphs, databases, APIs, plus text



03.

TYPES OF VECTOR SEARCH



WHAT IS VECTOR SEARCH?

What is Vector Search?

Traditional Search: Matches exact keywords (e.g., using SQL or Elasticsearch).

Vector Search: Matches meaning, not just words, using dense embeddings (numerical representations of text).

 **How It Works in RAG:** Embedding: The query and all documents are converted into vectors using a model (like Modern BERT). Similarity Comparison: The query vector is compared with document vectors using cosine similarity or dot product. Top Matches Retrieved: The most similar documents are passed to the generator to help create an informed answer.

 **Why Vector Search Matters in RAG:** Captures semantic meaning (e.g., understands synonyms and context) Improves relevance of retrieved content Enables better answers by providing the generator with higher-quality input



EXACT NEAREST NEIGHBOR (ENN) SEARCH

Exact Nearest Neighbor (ENN) Search is a search technique used to find the closest data points (neighbors) to a given query point in high-dimensional space. It compares the distances between vectors and retrieves the exact data points with the smallest distance to the query.



APPROXIMATE NEARESTNEIGHBOR (ANN) SEARCH

Approximate Nearest Neighbor (ANN) Search is a technique used to quickly find vectors (or data points) that are closest in meaning or distance to a given query vector—without having to scan the entire dataset exactly.



SEMANTIC SEARCH

Semantic search is a search technique that focuses on the meaning and context of a query, rather than just matching exact keywords.



SPARSE VECTOR SEARCH

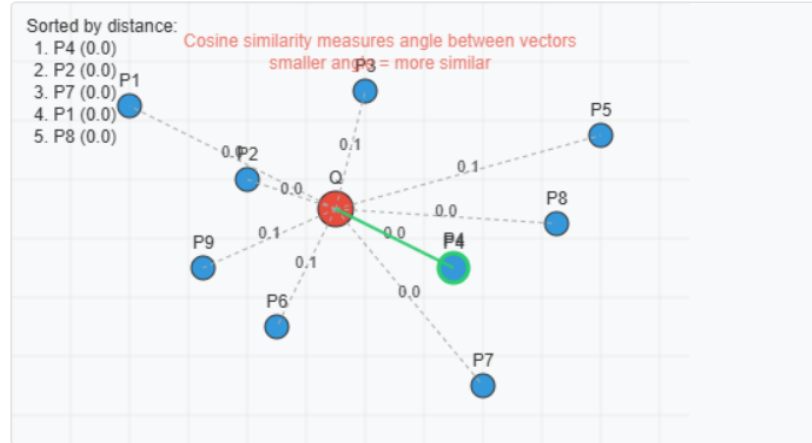
Sparse vector search is a traditional information retrieval method based on sparse (mostly zero) vector representations, like those used in TF-IDF or BM25 models.

Vector Search Simulation



1. Exact Nearest Neighbor Search (ENN)

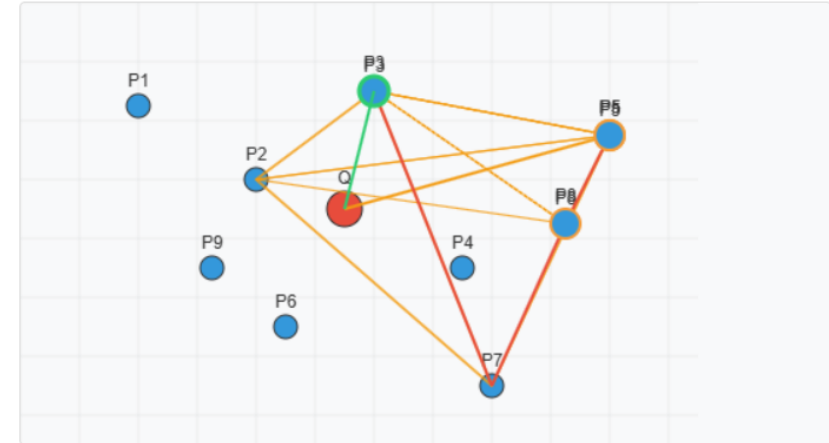
Finds the **exact** closest data points to a query by calculating distances to all vectors in the dataset.



Distance Metric: Cosine Similarity Step: 3. Return nearest neighbors

2. Approximate Nearest Neighbor Search (ANN)

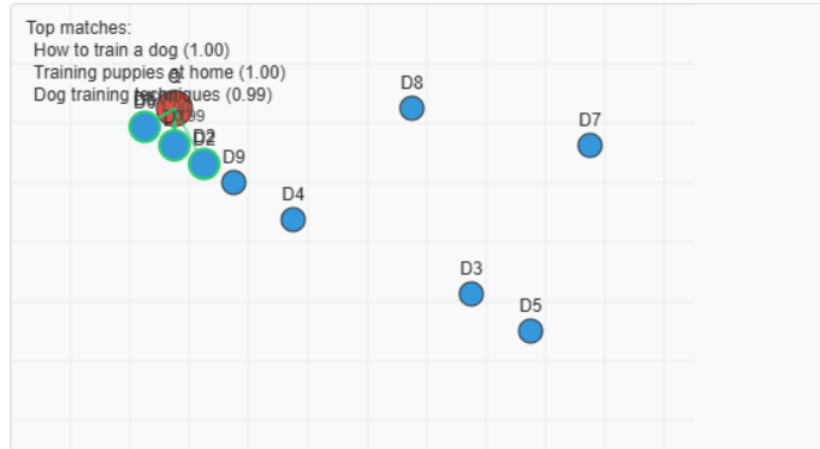
Sacrifices perfect accuracy for **speed** by using efficient data structures to approximate nearest neighbors.



Algorithm: Hierarchical NSW Step: 3. Return approximate NN

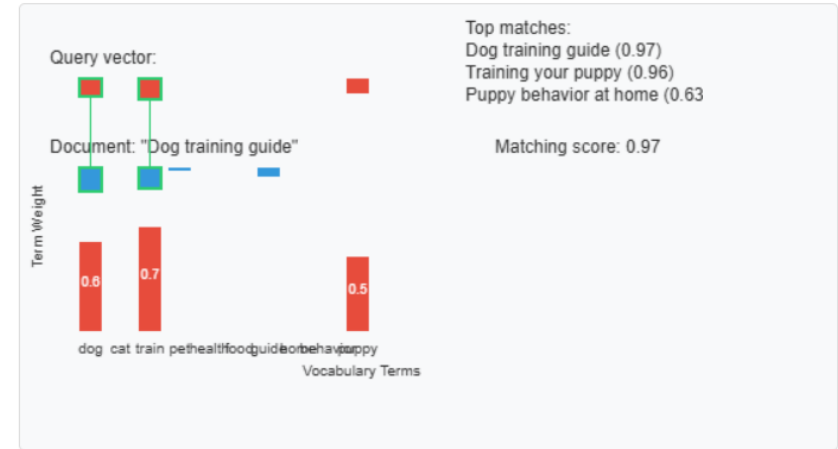
3. Semantic Search

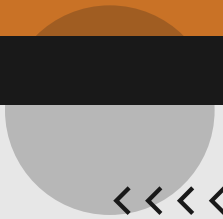
Uses **meaning** of content rather than keywords by searching through dense embedding vectors that capture semantic relationships.



4. Sparse Vector Search

Uses **high-dimensional sparse vectors** where most elements are zero, optimized for keyword and token matching.





04.

AGENTS



What is an Agent?



While conventional software enables users to streamline and automate workflows, agents are able to perform the same workflows on the users' behalf with a high degree of independence.

Agents are systems that **independently** accomplish tasks on your behalf.

A workflow is a sequence of steps that must be executed to meet the user's goal, whether that's resolving a customer service issue, booking a restaurant reservation, committing a code change, or generating a report.

Applications that integrate LLMs but don't use them to control workflow execution—think simple chatbots, single-turn LLMs, or sentiment classifiers—are not agents.

More concretely, an agent possesses core characteristics that allow it to act reliably and consistently on behalf of a user:

AGENT DESIGN FOUNDATION

In its most fundamental form, an agent consists of three core components:

- | | | |
|----|---------------------|---|
| 01 | Model | The LLM powering the agent's reasoning and decision-making |
| 02 | Tools | External functions or APIs the agent can use to take action |
| 03 | Instructions | Explicit guidelines and guardrails defining how the agent behaves |

HANDS-ON PRACTICE: HOW TO BUILD AGENTS?



OpenAI Agents SDK

**OpenAi Agents
SDK**

You can speak a bit
about this person here

<<<<



LangChain

You can speak a bit
about this person here

<<<<



SmolAgents
by
HuggingFace

smolagents

You can speak a bit
about this person here

<<<<

SMOLAGENTS LOOP

AL
EN
AD

AGENTIC AI

Main Points:

- Agentic AI refers to systems that exhibit **goal-directed behavior** and can act autonomously to achieve objectives.
- Contrasts with **tool-based AI**, which performs tasks reactively (e.g., ChatGPT without autonomy).
- Core properties:
 - **Autonomy**: Operates without continuous human input.
 - **Goal orientation**: Pursues objectives over time.
 - **Interactivity**: Engages with the environment to learn and adapt.

COMPONENTS OF AGENTIC AI

Main Points:

- **Perception:** Sensing and interpreting the environment.
- **Planning:** Strategizing actions to meet objectives.
- **Memory:** Retaining information for long-term coherence.
- **Action/Execution:** Carrying out decisions in the real world or digital environments.
- **Feedback loops:** Learning from outcomes to refine future behavior.

• Feedback loops: Learning from outcomes to refine future behavior.

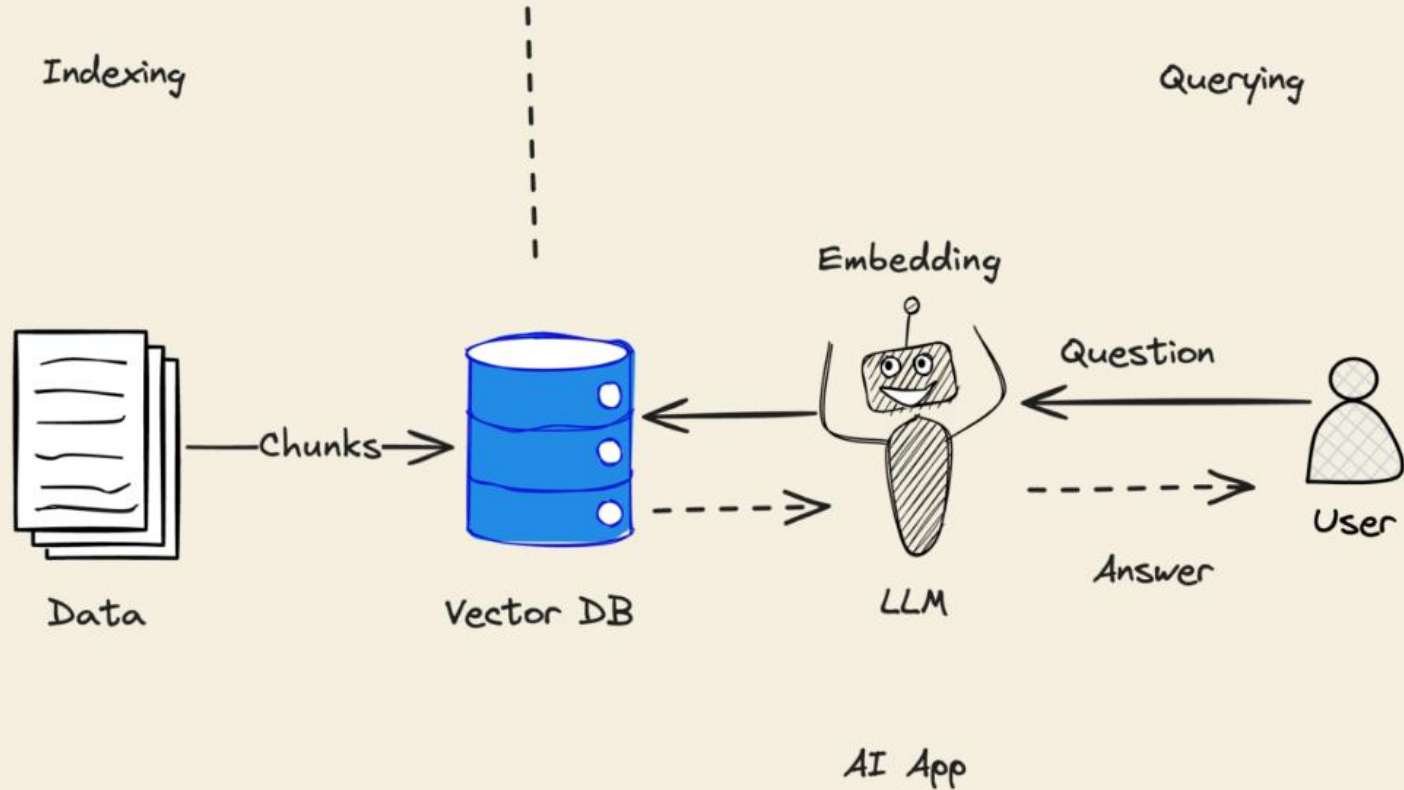


05.

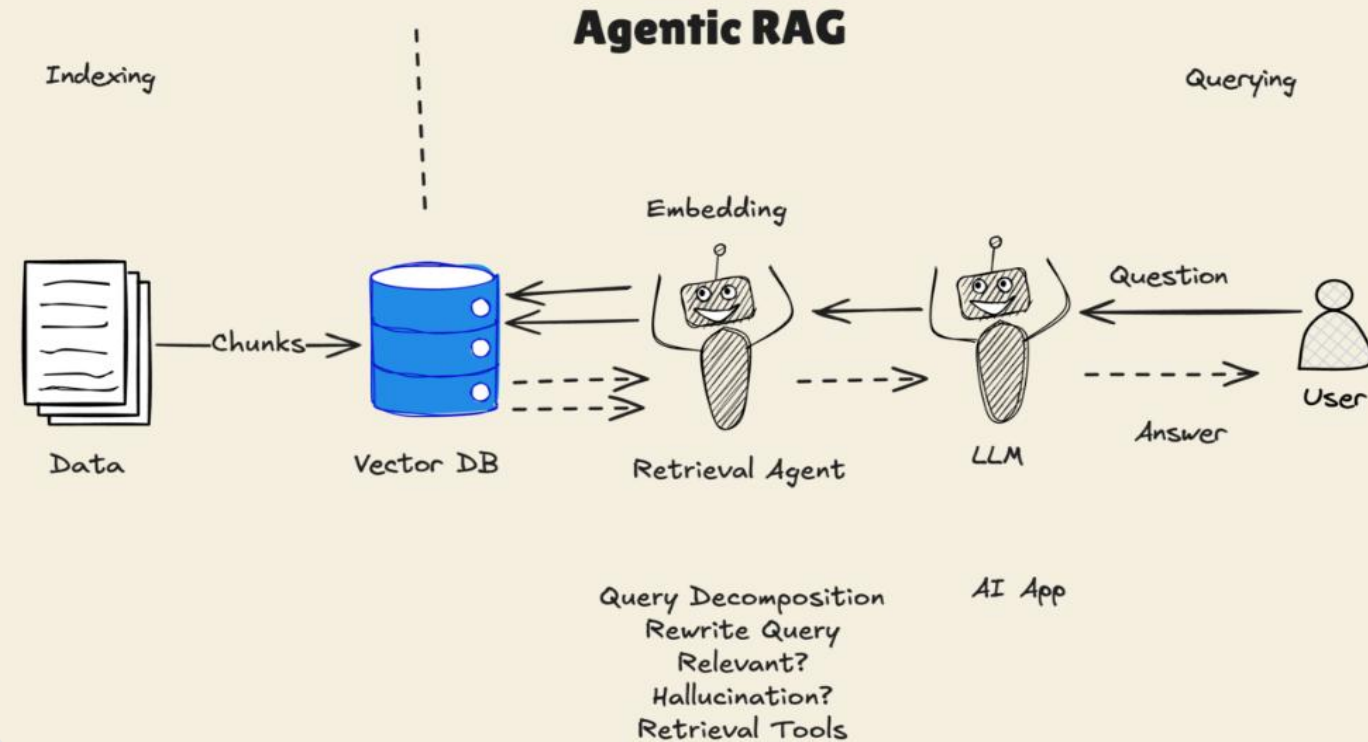
ADVANCED AGENT LIBRARIES AND TOOLS



BEFORE START REMEMBER!



AGENTIC RAG



ARTIFICIAL AGNO

INTELLIGENCE [AI]



- ❑ Agno is a lightweight library for building Agents with memory, knowledge, tools and reasoning
- ❑ Developers use Agno to build Reasoning Agents, Multimodal Agents, Teams of Agents and AgenticAI Workflows. Agno also provides a beautiful UI to chat with your Agents, pre-built FastAPI routes to serve your Agents and tools to monitor and evaluate their performance.
- ❑ Click on Notebook to make experience.



[AI]

DEEPFLOW

DeepFlow[®]

Instant Observability for Cloud & AI Applications

Zero Code, Full Stack, eBPF & Wasm

DOI 10.5281/zenodo.15303230 release v7.0.7  [docker pulls](#) license Apache-2.0

CRAWL4AI

Crawl4AI: Open-source LLM Friendly Web Crawler & Scraper.



GITHUB TRENDING

#1 Repository Of The Day



Stars

43k



Forks

4k

pypi package 0.6.3

python 3.9 | 3.10 | 3.11 | 3.12 | 3.13

downloads/month 486k

license Apache-2.0

code style black

security bandit

Contributor Covenant 2.1



Star

3,281

The Open Source Platform for Training Advanced AI Models

With Transformer Lab, researchers, ML engineers, and developers can collaborate to build, study, and evaluate AI models—with provenance, reproducibility, evals, and transparency included.



Prompt-friendly codebase



Turn any Git repository into a simple text digest of its codebase.

This is useful for feeding a codebase into any LLM.

This is useful for feeding a codebase into any LLM.

Turn any Git repository into a simple text digest of its codebase.

SYNTHETIC DATA KIT



meta-llama/**synthetic-data-kit**



Tool for generating high quality Synthetic datasets



3

Contributors



2

Used by



774

Stars



78

Forks



MEMO



mem0

The Memory Layer
for Personalized AI



GITHUB TRENDING

#1 Repository Of The Day

[Learn more](#) · [Join Discord](#) · [Demo](#) · [OpenMemory](#)

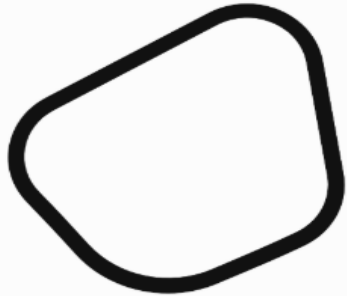
Mem0 4027 members · downloads 1M/month · commit activity 62/month · pypi package v0.1.98 · npm v2.1.25 · Y Combinator S24

[Building Production-Ready AI Agents with Scalable Long-Term Memory →](#)

⚡ +26% Accuracy vs. OpenAI Memory • 🚀 91% Faster • 💰 90% Fewer Tokens

⚡ +26% Accuracy vs. OpenAI Memory • 🚀 91% Faster • 💰 90% Fewer Tokens

MORPHIK



morphik

PRs welcome commit activity 98/month issues 26 closed downloads 872/month discord 27 online

[Docs](#) - [Community](#) - [Why Morphik?](#) - [Bug reports](#)

Morphik is an alternative to traditional RAG for highly technical and visual documents.



rowboat

Let AI build multi-agent workflows for you in minutes

[Quickstart](#) | [Docs](#) | [Discord](#) | [Website](#) | [Youtube](#)

- ✨ **Start from an idea** -> copilot builds your multi-agent workflows
 - E.g. "Build me an assistant for a food delivery company to handle delivery status and missing items. Include the necessary tools."
- 🌐 **Connect MCP servers**
 - Add the MCP servers in settings -> import the tools into Rowboat.
- 📞 **Integrate into your app using the HTTP API or Python SDK**
 - Grab the project ID and generated API key from settings and use the API.

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06.

MODEL CONTEXT PROTOCOL (MCP)



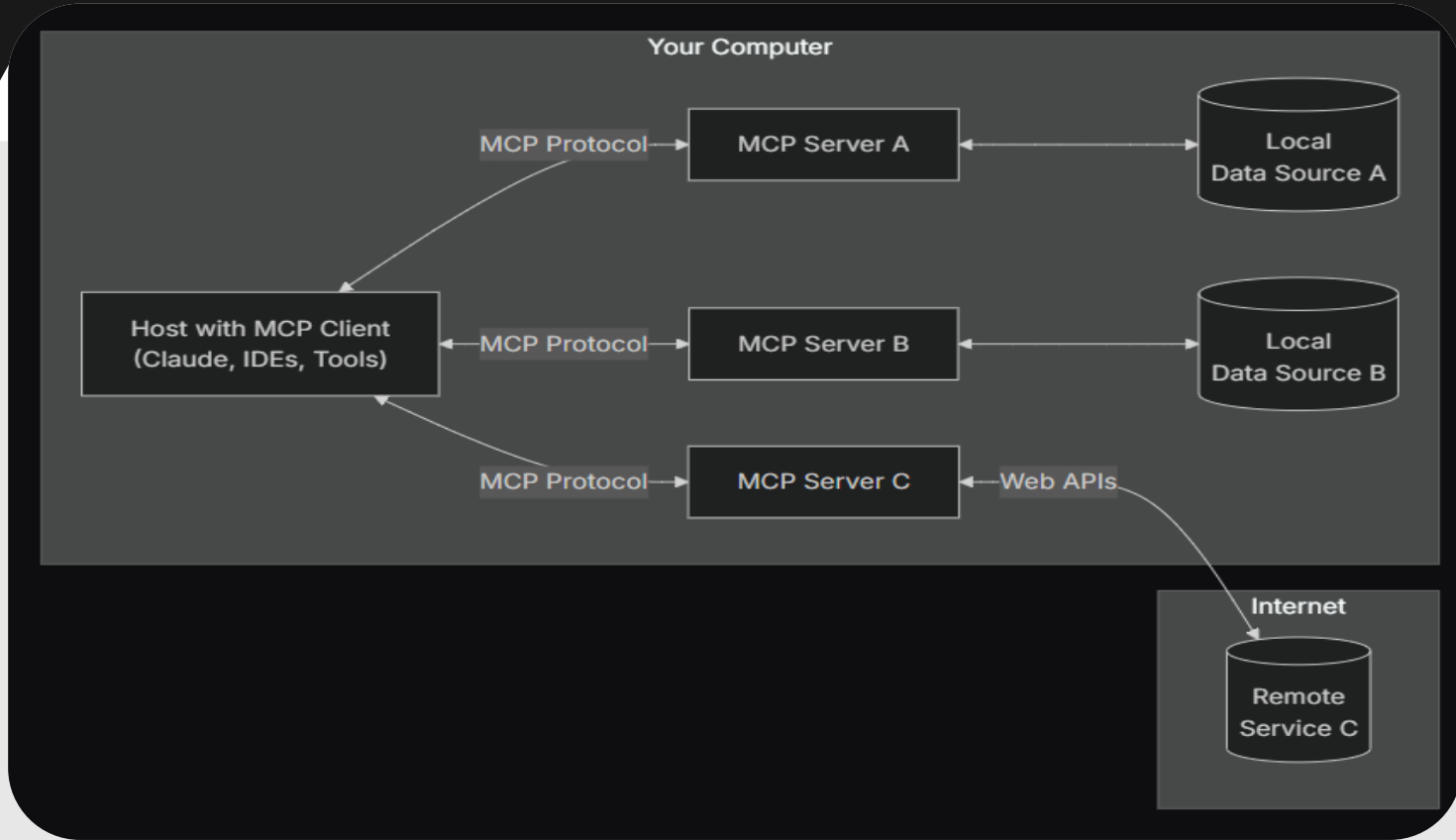
WHY MCP?



MCP helps you build agents and complex workflows on top of LLMs. LLMs frequently need to integrate with data and tools, and MCP provides:

- A growing list of pre-built integrations that your LLM can directly plug into
- The flexibility to switch between LLM providers and vendors
- Best practices for securing your data within your infrastructure

MCP ARCHITECTURE

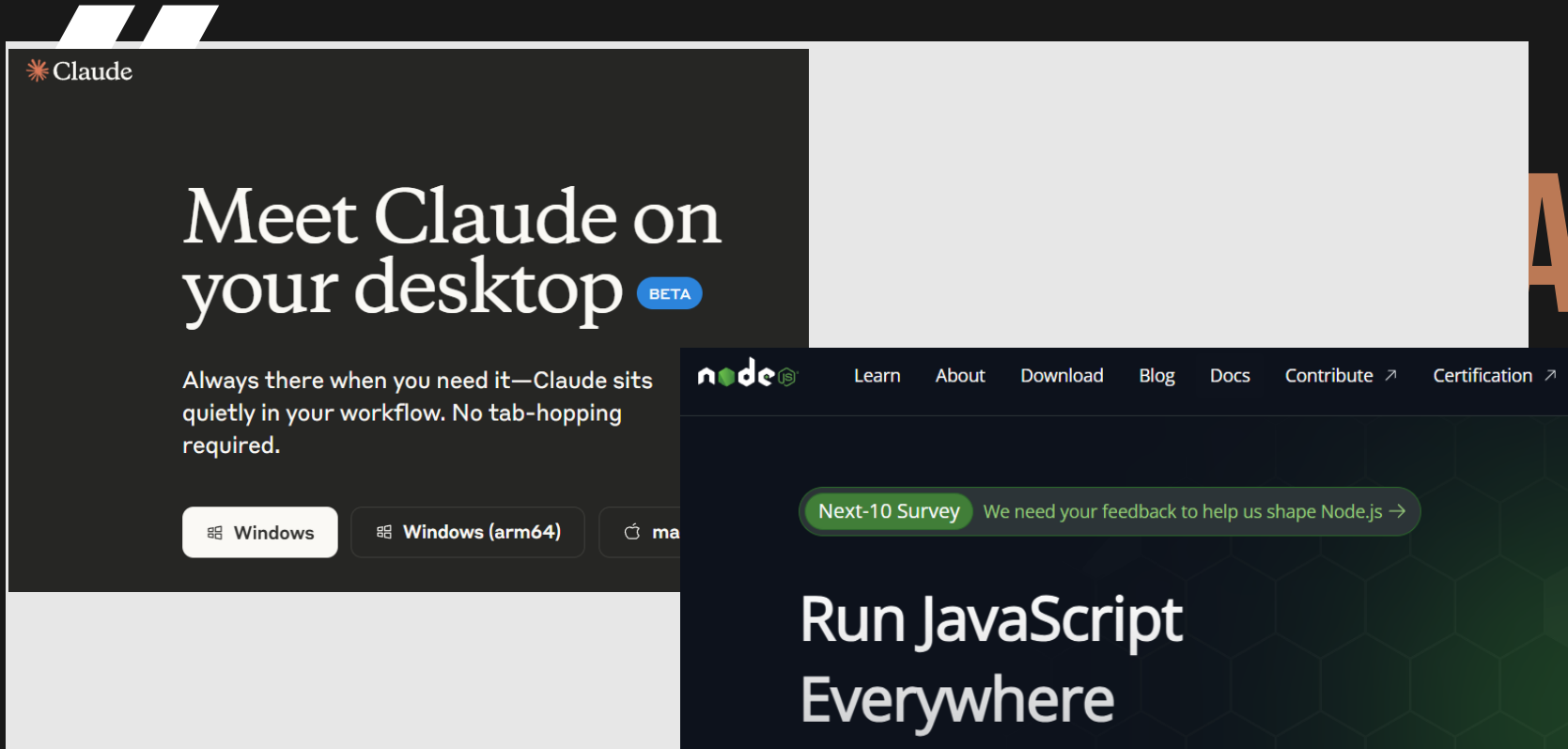


- **MCP Hosts:** Programs like Claude Desktop, IDEs, or AI tools that want to access data through MCP
- **MCP Clients:** Protocol clients that maintain 1:1 connections with servers
- **MCP Servers:** Lightweight programs that each expose specific capabilities through the standardized Model Context Protocol
- **Local Data Sources:** Your computer's files, databases, and services that MCP servers can securely access
- **Remote Services:** External systems available over the internet (e.g., through APIs) that MCP servers can connect to

that MCP servers can connect to

- **Remote Services:** External systems available over the internet (e.g., through APIs)

HOW TO WORK WITH MCP?



DEFINE OUR FIRST MCP SERVER



Model Context Protocol

Project Structure

- [specification](#) - Protocol specification and documentation
- [typescript-sdk](#) - TypeScript implementation
- [python-sdk](#) - Python implementation
- [java-sdk](#) - Java implementation
- [kotlin-sdk](#) - Kotlin implementation
- [csharp-sdk](#) - C# implementation
- [docs](#) - User documentation and guides
- [create-kotlin-server](#) - Kotlin sample server
- [servers](#) - List of maintained servers

MCP FILESYSTEM

P list the all files that i have

I'll help you list all the files you have. Let me check the available directories first.

L list_allowed_directories

Now I'll list all the files in this directory:

L list_directory

You have 3 files in your directory:

1. Slides.pdf
2. Slides2.pdf
3. Slides3.pptx

These are all located in C:\Users\Pejman\OneDrive\Desktop\PPT_FL. Let me know if you need more information about any of these files.

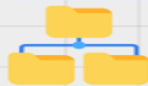


Retry

Claude can make mistakes. Please double-check responses

12 MCP Servers You Can Use in 2025

1 File System Server



Gives the LLM direct access to the local file system to read, write, create directories

2 GitHub MCP Server



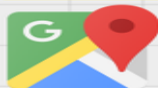
Connects Claude to GitHub repos and allows file updates, code searching

3 Slack MCP Server



MCP Server for Slack API, enabling Claude to interact with Slack workspaces

4 Google Maps MCP Server



MCP Server for Google Maps API

5 Docker MCP Server



Integrate with Docker to manage containers, images, volumes, and networks

6 Brave MCP Server



Web and local search using Brave's Search API

7 PostgreSQL MCP Server



A MCP Server that enables LLM to inspect database schemas and execute read-only queries

8 Google Drive MCP Server



MCP Server integrates with Google Drive to allow reading and searching over files

9 Redis MCP Server



MCP Server that provides access to Redis databases

10 Notion MCP Server



This project implements an MCP Server for the Notion API

11 Stripe MCP Server



MCP Server to interact with the Stripe API

12 Perplexity MCP Server

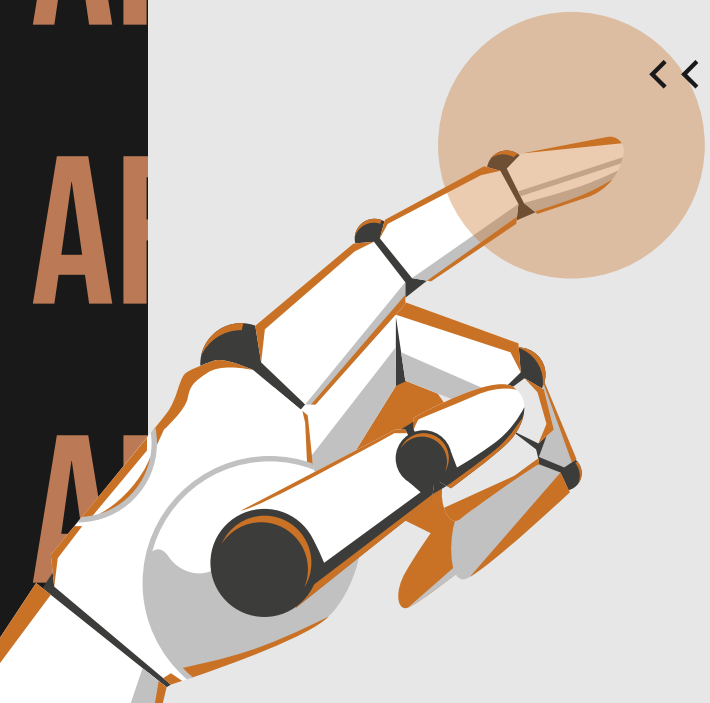
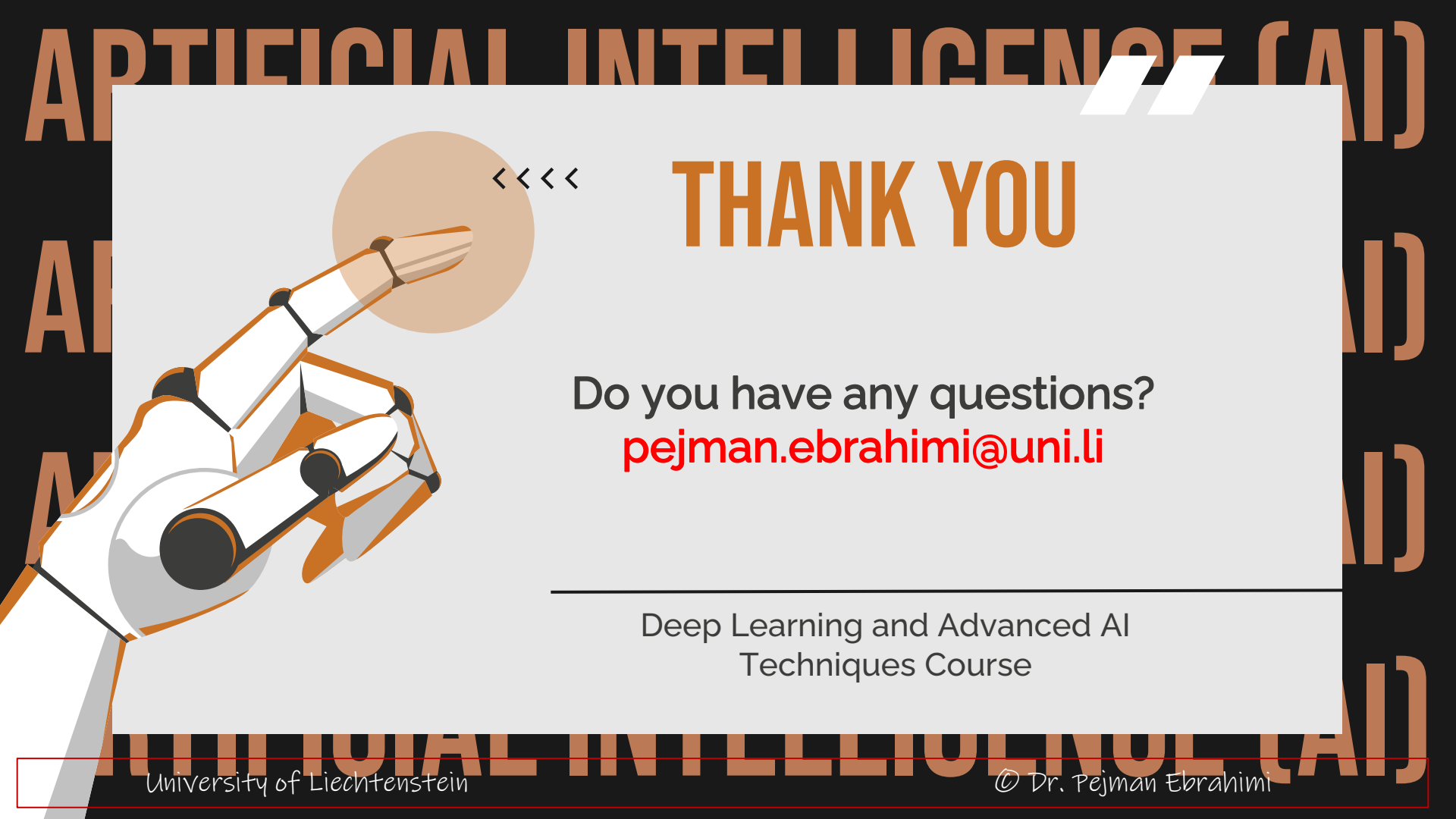


An MCP Server that connects to Perplexity's Sonar API for real-time search

MCP A2A AG-UI

The Agent Protocol Stack





<<<<

THANK YOU

Do you have any questions?

pejman.ebrahimi@uni.li

Deep Learning and Advanced AI
Techniques Course